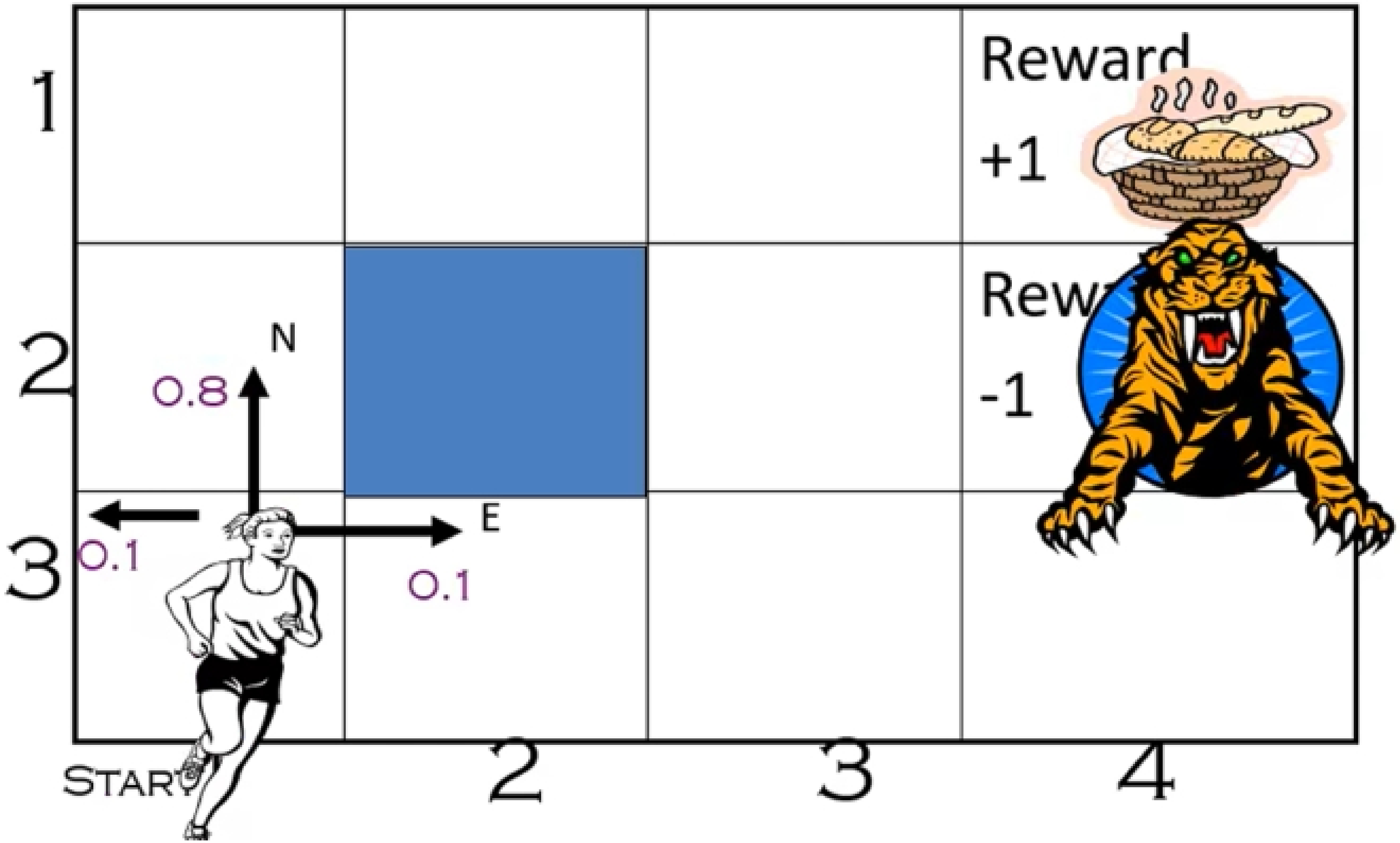
# Markov Decision Process (MDP)
# Chapter 17: Making Complex Decisions

- Defined as a tuple: <S, A, P, R>
  - **S:** State
  - **A:** Action
  - **P:** Transition function
    - Table P(s'| s, a), prob of s' given action "a" in state "s"
  - **R:** Reward
    - R(s, a) = cost or reward of taking action a in state s

- Choose a sequence of actions (not just one action)
  - Utility based on a sequence of actions
  - Model **Sequential Decision Problems**

# Example: What SEQUENCE of actions should our agent take?

- Agent can take action N, E, S, W
- Each action costs –1/25



# MDP Tuple: <S, A, P, R>

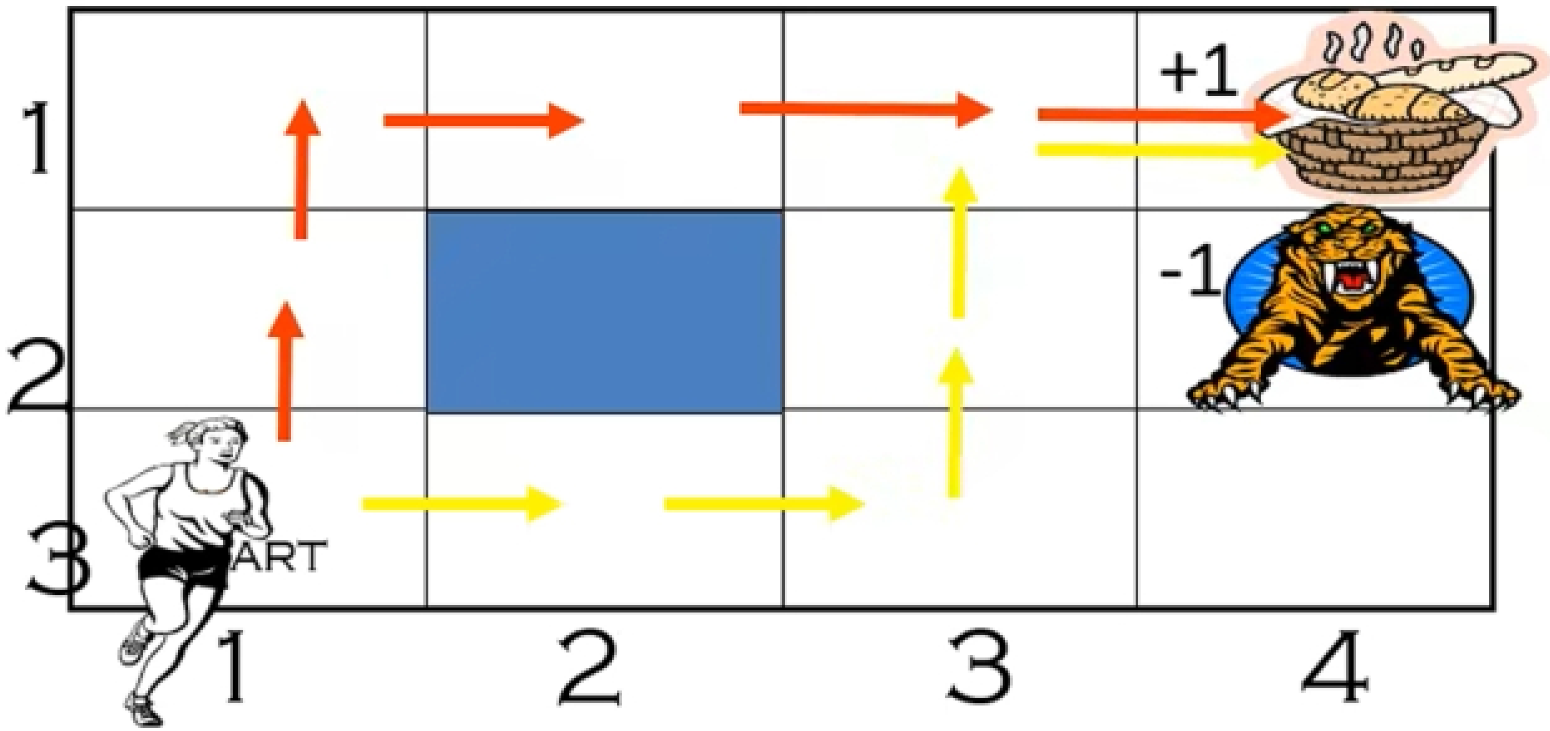
- **S:** State of the agent on the grid
  - Ex: state (4,3)

- **A:** Actions of the agent, i.e., N, E, S, W

- **P:** Transition function
  - Table $P(s' | s, a)$, prob of s' given action "a" in state "s"
  - E.g., $P( (4,3) \mid (3,3), N) = 0.1$
  - E.g., $P((3, 2) \mid (3,3), N) = 0.8$
  - (Robot movement, uncertainty of another agent's actions,…)

# How Would you Solve this Problem?

- Simple search algorithm? **Not deterministic**
- Apply MEU to an entire sequence of actions?
  - *Create multiple plans, e.g., Red plan vs. Yellow plan below*
  - *Choose a plan that leads to MEU*



# How Would you Solve this Problem?

- Apply **MEU** to an entire sequence of actions?
- Does not work because uncertainty at every step
  - E.g., After first step of Red plan, may move east not North!
  - No action specified there (I.e., in cell (2,1))

- Solution is a *Policy*
  - Complete mapping from states to actions

# MDP Basics and Terminology

- **Markov Assumption:** Transition probabilities (and rewards) from any given state depend only on the state and not on previous history

- An agent must make a decision or control a probabilistic system
  - Goal is to choose a sequence of actions for optimality
  - **Decision Epoch:** Points at which decisions are made
    - **Finite horizon MDPs**: # of decision epochs is finite i.e. fixed time after which game ends : Time dependent policy
    - **Infinite horizon MDPs:** # of decision epochs is infinite i.e. Time independent policy
  - **Transition model:** Table of probabilities P
    - In our example, 0.8, 0.1, 0.1 transition probabilities
    - P(J | S, A) : Probability of state J, given action A in State S
  - **Absorbing state:** Goal state

# Reward Function

- Reward is assumed associated with state, action i.e. **R(S, A)**
  - If all actions have the same reward can use R(S)
  - We could also assume a mix of R(S,A) and R(S)
  - Will use R(S,A) as the notation

- Sometimes, reward associated with state, action, destination-state
  - R(S,A,J)
  - $R(S,A) = \sum_{J} R(S,A,J) * P(J \mid S, A)$

# MDP Policy

- **Decision Rule:** Procedure to choose action in each state for ***a given decision epoch***
  - E.g., MDP has states, S1 and S2, with actions A1, A2 in both states
  - Decision rules Di for each decision epoch "i" as shown in table below
  - Four decision rules shown, D1, D2, D3, D4, one for each epoch
  - Numbers in (..) are probabilities, e.g., 0.7, 0.3, 1.0
- **Policy:** Decision rule to be used at all decision epochs
  - Policy = {D1, D2, D3,D4} (assuming finite horizon T = 4)

| D1 | D2 | D3 | D4… |
|---|---|---|---|
| S1→A1 (0.7)<br>→A2 (0.3)<br>S2→A2 (1.0) | S1→A1 (1.0)<br>S2→A1 (0.3)<br>→A2 (0.7) | S1→A2 (1.0)<br>S2→A2 (1.0) | …. |

# Stationary and Deterministic Policies

- **Stationary** policy implies same decision rule in every epoch
  - **Stationary policy:** {D, D, D, D…}
  - **Non-stationary policy** changes with time (e.g., D1,D2, D3…Dn)
- **Deterministic policy** implies choosing an action with certainty
  - **Deterministic policy:** Si → Ai (probability 1.0)
  - **Randomized policy:** Probability distribution on the set of actions
- What type of a policy is the following?

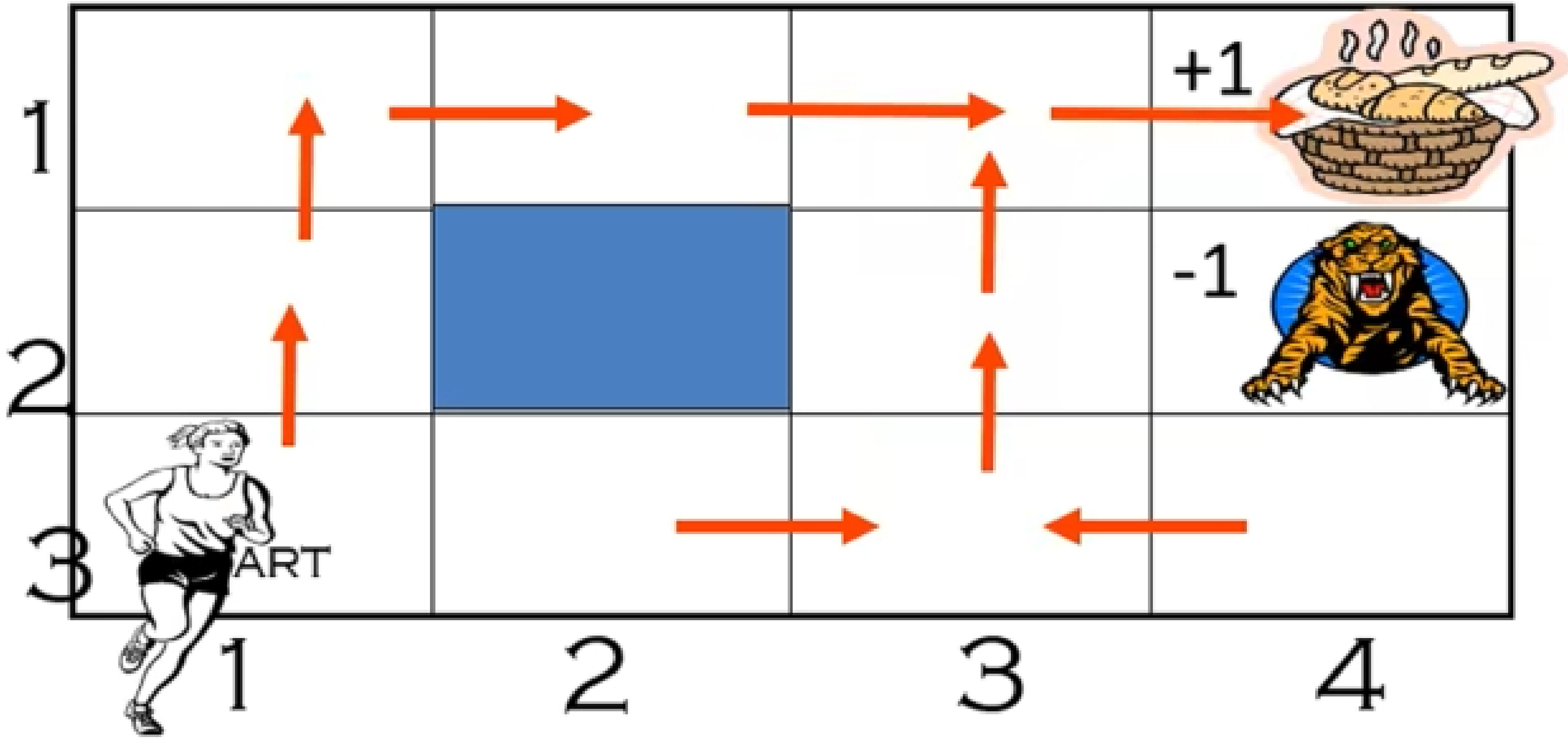
| D | D | D | D |
|---|---|---|---|
| S1→A1 (1.0)<br>S2→A2 (1.0) | S1→A1 (1.0)<br>S2→A2 (1.0) | S1→A1 (1.0)<br>S2→A2 (1.0) | S1→ A1(1.0)<br>S2→ A2(1.0) |

# Stationary and Deterministic Policies

- **Optimal** MDP policy for infinite horizon is Stationary & Deterministic policies (aka pure policy)
- Policy denoted by symbol $\pi$
- Stationary & deterministic policies denoted $\pi^{SD}$
- Is a policy $\pi^{SR}$ possible? (SR = Stationary & randomized)

- **Note:**
  - **When nothing is specified regarding time horizon, assume infinite horizon**
  - **When asked to find the policy at time horizon = 4, it means find the decision rule D4. It can also be stated as find decision rule for T = 4 or D4.**
  - **When asked to find policy for a time horizon of 4, means find all decision rules D1, D2, D3 and D4**

# Pure Policies: $\pi^{SD}$




- Deterministic, non-changing mapping from states to actions

  $\pi((1,3)) \rightarrow$ North (non-changing, non-random)

  $\pi((1,2)) \rightarrow$ North

  $\pi((4,3)) \rightarrow$ West……

# Policy

- **Policy** is like a plan, but not quite
  - Certainly, generated ahead of time, like a plan
- Unlike traditional plans, it is not a sequence of actions that an agent must execute
  - If there are failures in execution, agent can continue to execute a policy
- Prescribes an action for all the states
- Maximizes expected reward, rather than just reaching a goal state